

Simplifying Buffer Placement MILP Using Precomputed Throughput

1 Introduction

The buffer placement MILP model includes two types of constraints: timing constraints, which enforce buffers to break signal paths on certain channels, and throughput constraints, which determine whether additional buffers are needed to accommodate tokens for achieving maximum throughput.

Circuit throughput appears in the throughput constraints and serves as the optimization objective. We observe that the maximum throughput can be precomputed by solving only the timing constraints on data & valid signal paths. This value can then be substituted into the throughput constraints, simplifying the model, especially when treating throughput as a variable introduces quadratic terms.

This approach relies on the assumption that buffer timing delays are not modeled. That is, once the timing constraints are satisfied, adding buffers will not violate them afterward.

2 Model Description

The MILP models are essentially formulated as follows.

Assumptions:

- **Single CFC:** All the buffer placement MILPs operate on MILP variables and constants derived from *choice-free circuits* (CFCs).
- **Single buffer:** Each channel has up to one buffer (only one type).

Notations:

- A choice free circuit has a set of units ($u \in U$) and a set of channels ($c \in C$).

Table 1 lists all the constants and variables used in different MILP models.

2.1 Timing Constraints

Target Period. For each channel c :

$$T_c^{D,out} \leq \text{target_period}. \quad (1)$$

$$T_c^{V,out} \leq \text{target_period}. \quad (2)$$

$$T_c^{R,out} \leq \text{target_period}. \quad (3)$$

$$T_c^{D,in} \leq \text{target_period}. \quad (4)$$

$$T_c^{V,in} \leq \text{target_period}. \quad (5)$$

$$T_c^{R,in} \leq \text{target_period}. \quad (6)$$

Input parameters:		
target_period	$\mathbb{R}_{\geq 0}$	Target clock period.
backedge_c	$\mathbb{B}$	Whether c is a backedge.
latency_u	$\mathbb{N}$	The latency of unit.
delay_u^{Sig₀→Sig₁}	$\mathbb{R}_{\geq 0}$	u 's delay (domains $\text{Sig}_0 \rightarrow \text{Sig}_1$).
Output variables:		
num_slots_c	$\mathbb{N}$	Number of buffer slots on c .
buffered_c^D	$\mathbb{B}$	If the data signal of c is broken.
buffered_c^V	$\mathbb{B}$	If the valid signal of c is broken.
buffered_c^R	$\mathbb{B}$	If the ready signal of c is broken.
Internal variables:		
<i>Timing variables:</i>		
T_c^{D,in}	$\mathbb{R}_{\geq 0}$	Data arrival time at c 's input.
T_c^{D,out}	$\mathbb{R}_{\geq 0}$	Data arrival time at c 's output.
T_c^{V,in}	$\mathbb{R}_{\geq 0}$	Valid arrival time at c 's input.
T_c^{V,out}	$\mathbb{R}_{\geq 0}$	Valid arrival time at c 's output.
T_c^{R,in}	$\mathbb{R}_{\geq 0}$	Ready arrival time at c 's input.
T_c^{R,out}	$\mathbb{R}_{\geq 0}$	Ready arrival time at c 's output.
<i>Throughput variables:</i>		
throughput	$\mathbb{R}_{\geq 0}$	CFC's throughput.
tokens_c	$\mathbb{R}_{\geq 0}$	c 's token count.
bubbles_c	$\mathbb{R}_{\geq 0}$	c 's bubble count.
retiming_u	$\mathbb{R}_{\geq 0}$	Token retiming over u .

Table 1. MILP Variables

Channel delay propagation. For each channel c :

$$T_c^{D,out} \geq T_c^{D,in} - \text{buffered}_c^D \cdot \text{target_period}. \quad (7)$$

$$T_c^{V,out} \geq T_c^{V,in} - \text{buffered}_c^V \cdot \text{target_period}. \quad (8)$$

$$T_c^{R,out} \geq T_c^{R,in} - \text{buffered}_c^R \cdot \text{target_period}. \quad (9)$$

$$\text{buffered}_c^D = \text{buffered}_c^V. \quad (10)$$

Unit delay propagation. For each pair of input channel c_i and output channel c_j (c_i and c_j can be the same channel) of u , consider all cases of input/output signals:

$$T_{c_j}^{sig_0,in} \geq T_{c_i}^{sig_1,out} + \text{delay}_u^{Sig_0 \rightarrow Sig_1}. \quad (11)$$

2.2 Throughput Constraints

Reachability of the steady state. This constraint ensures that the steady state that the MILP tries to optimize is reachable. For each channel c connects u and v ,

$$\text{tokens}_c = \text{backedge}_c + \text{retiming}_v - \text{retiming}_u. \quad (12)$$

Channel throughput. These constraints ensure that the buffer type can support the number of tokens and bubbles in the reachable state for a certain throughput. For each channel c :

$$\text{throughput} \leq \text{tokens}_c - \text{buffered}_c^D + 1 \quad (13)$$

$$\text{throughput} \leq \text{bubbles}_c - \text{buffered}_c^R + 1 \quad (14)$$

$$\text{bubbles}_c + \text{tokens}_c \leq \text{num_slots}_c. \quad (15)$$

Pipelined unit throughput. This constraint ensures that each pipelined unit works well under a certain throughput.

$$\text{latency}_u \cdot \text{throughput} = \text{retiming}_{u_out} - \text{retiming}_{u_in} \quad (16)$$

2.3 Objective

Maximum throughput & minimum buffer slots. Throughput is the highest priority, and λ is a sufficiently small coefficient.

$$\max. \quad \text{throughput} - \lambda \cdot \sum_c \text{num_slots}_c. \quad (17)$$

3 Simplification of the Model

We simplify the MILP in two steps.

3.1 Step 1: Compute the Maximum Throughput

In the first step, we apply timing constraints on data and valid signal paths:

$$T_c^{D,out} \leq \text{target_period}. \quad (18)$$

$$T_c^{V,out} \leq \text{target_period}. \quad (19)$$

$$T_c^{D,in} \leq \text{target_period}. \quad (20)$$

$$T_c^{V,in} \leq \text{target_period}. \quad (21)$$

$$T_c^{D,out} \geq T_c^{D,in} - \text{buffered}_c^D \cdot \text{target_period}. \quad (22)$$

$$T_c^{V,out} \geq T_c^{V,in} - \text{buffered}_c^V \cdot \text{target_period}. \quad (23)$$

$$\text{buffered}_c^D = \text{buffered}_c^V. \quad (24)$$

$$T_{c_j}^{sig_0,in} \geq T_{c_i}^{sig_1,out} + \text{delay}_u^{\text{Sig}_0 \rightarrow \text{Sig}_1}. \quad (25)$$

We then minimize the number of signal breaks (where buffered_c^D is 1) on the cycle that contains the most such breaks. This number corresponds to the minimum initiation interval (Π), whose reciprocal gives the maximum throughput.

To implement this, we introduce integer variables countValid_u for each unit u , which accumulate the number of signal breaks along paths. For a non-backed channel connecting two units:

$$\text{countValid}_{u,succ} - \text{countValid}_{u,pred} \geq \text{buffered}_c^D. \quad (26)$$

For backed channels, we measure the total number of breaks along their associated cycles using the following inequality:

$$\Pi \geq \text{countValid}_{u,pred} - \text{countValid}_{u,succ} + \text{buffered}_c^D. \quad (27)$$

The left-hand side Π represents the initiation interval, and the right-hand side counts the total number of signal breaks ($\text{buffered}_c^D = 1$) in the cycle. Therefore, the objective is:

$$\min \Pi. \quad (28)$$

3.2 Step 2: Solve the Full Model

After computing the maximum throughput (i.e., $1/\Pi$), we substitute it as a constant into the throughput constraints. This simplifies the model—especially when throughput appears in nonlinear terms. Although this basic example remains linear, models with quadratic terms containing throughput benefit significantly from this approach.

We still need to resolve the full model, including the timing constraints, since the specific placement of signal breaks can affect the whole model, even if their total count on a cycle remains fixed.

3.3 Justification

Once the maximum throughput is known, we can add extra buffers to satisfy the remaining constraints without degrading it. For instance, TEHBs on all channels resolve timing on ready paths, and transparent FIFOs with infinite slots ensure throughput feasibility. As the model does not account for buffer delay, these additions preserve timing correctness.

A formal proof requires certain structural assumptions, which generally hold for dataflow circuits.